

45-Day Cyber Security & Ethical Hacking Course

Day-Wise Syllabus

Module 1 — Introduction to Cyber Security & Hacking

Day 1

- Introduction to Cyber Security
- Importance of Cyber Security
- Real-world Cyber Attacks

Day 2

- What is Hacking
- Types of Hackers
 - White Hat
 - Black Hat
 - Grey Hat

Day 3

- Phases of Hacking
- How Hackers Think
- How Hackers Target Their Victims

Day 4

- Cyber Laws & Ethics
- Password Security Awareness
- Quiz & Discussion

Module 2 — Linux, Kali Linux & Command Line

Day 5

- What is Command Line and GUI
- Introduction to Linux
- Linux Distributions

Day 6

- Introduction to Kali Linux
- Installing Kali Linux
- VirtualBox Setup

Day 7

- Basic Linux Commands
 - ls
 - pwd
 - cd
 - mkdir
 - rm

Day 8

- More Linux Commands
 - cat
 - touch
 - cp
 - mv
 - clear
- File Permissions

Day 9

- User Management
- Package Installation
- Linux Practical Session

Module 3 — Phishing & Social Engineering

Day 10

- Introduction to Phishing
- Types of Phishing Attacks

Day 11

- Email Phishing
- Fake Login Pages
- SMS & Social Media Scams

Day 12

- Introduction to Social Engineering
- Human Psychology in Hacking

Day 13

- Identity Theft
- Online Fraud Awareness
- Social Media Security

Day 14

- Phishing Awareness Practical
- Identifying Fake Websites & Emails
- Quiz

Module 4 — Networking Fundamentals

Day 15

- Introduction to Networking
- Types of Networks
 - LAN
 - WAN
 - MAN

Day 16

- IP Address
- Public vs Private IP
- MAC Address

Day 17

- Ports & Protocols
- TCP/IP
- HTTP, HTTPS, FTP, SSH

Day 18

- DNS (Domain Name System)
- What is a Server
- How Websites Work

Day 19

- Firewall Basics
- VPN & Proxy
- Networking Practical

Module 5 — OSI Model & Layer-wise Attacks

Day 20

- Introduction to OSI Model
- 7 Layers of OSI

Day 21

- Layer-wise Attacks
- Packet Flow Understanding

Day 22

- Packet Analysis Practical using Wireshark

Module 6 — Footprinting & Reconnaissance

Day 23

- What is Footprinting
- Types of Footprinting
- Information Gathering

Day 24

- Using ping
- Using nslookup
- Whois Lookup

Day 25

- DNS Enumeration using dnsenum
- OSINT Basics
- Practical Session

Module 7 — Scanning & Enumeration

Day 26

- Introduction to Scanning
- Types of Scanning

Day 27

- Open, Closed, Filtered & Unfiltered Ports

- Banner Grabbing

Day 28

- Introduction to Nmap
- Network Scanning using Nmap

Day 29

- hping3 Basics
- Flag Scanning
- Zenmap Demonstration

Module 8 — Web Servers & Website Security

Day 30

- Introduction to Web Servers
- Website Security Basics

Day 31

- Web Server Vulnerabilities
- Admin Panel Security

Day 32

- File Upload Vulnerability
- Web Shell Concepts

Day 33

- Introduction to DVWA
- Installing XAMPP

Day 34

- Vulnerability Scanning using Nikto
- Web Server Scanning with Nmap

Module 9 — Sniffing & Traffic Analysis

Day 35

- What is Sniffing
- Packet Sniffing Basics

Day 36

- Traffic Analysis Practical using Wireshark

Module 10 — Password Security & Cracking Concepts

Day 37

- Password Security
- Dictionary Attacks
- Brute-force Concepts

Day 38

- Wordlist Creation
- Password Protection Techniques
- Practical Demonstration

Module 11 — Windows & Android Security

Day 39

- Introduction to Windows Security
- Windows Firewall
- Virtual Machines

Day 40

- Introduction to Android Security
- Mobile Security Awareness

Module 12 — DoS, DDoS, Dark Web & Anonymity

Day 41

- What is DoS & DDoS
- Types of DDoS Attacks
- Prevention Basics

Day 42

- Introduction to Dark Web
- VPN & Proxy
- Online Anonymity

Module 13 — Vulnerability Assessment, Cryptography & Steganography

Day 43

- Vulnerability Assessment
- Introduction to Cryptography
- Encryption & Hashing

Day 44

- Introduction to Steganography
- Hidden Data Concepts
- Practical Demonstration

Module 14 — Final Project & Assessment

Day 45

- Final Practical Assessment
- Student Project Presentation
- Viva / Oral Test
- Feedback Session

- Certificate Distribution

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